# On Biadjoint Triangles

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CONTENTS

# 1 MONADS

No self-respecting wannabe category theorist would leave out a description of monads. In this section, a comprehensive overview of monads will be given, with the Kleisli and Eilenberg-Moore categories defined in detail. Much of this comes directly from Mac Lane, and will begin with a quick remark on composeability of functors and natural transformations.

For a category $\mathbb{A}$, and endofunctor $T : \mathbb{A} \to \mathbb{A}$ there are composites $T^2 = T \circ T$ and $T^3 = T^2 \circ T$. If, additionally, there is a natural transformation $\mu : T^2 \to T$ then, for each $a$ in $\mathbb{A}$, the natural transformations $T\mu, \mu T : T^3 \to T^2$ have, respectively components $(T\mu)_a = T\mu_a$, and $(\mu T)_a = \mu_{Ta}$ .

**Definition 1.1.** [**MaclCT**] A *monad* $T = (T, \eta, \mu)$ in a category $\mathbb{A}$ consists of a functor $T : \mathbb{A} \longrightarrow \mathbb{A}$ and two natural transformations

$$\eta : 1_{\mathbb{A}} \longrightarrow T, \quad \mu : T^2 \longrightarrow T$$

such that the following diagrams commute:

$$\begin{tikzcd}
T^3 \arrow[r, "T\mu"] \arrow[d, "\mu T"'] & T^2 \arrow[d, "\mu"] \\
T^2 \arrow[r, "\mu"'] & T
\end{tikzcd}
\qquad
\begin{tikzcd}
1T \arrow[r, "\eta T"] & T^2 \arrow[d, "\mu"] & T1 \arrow[l, "T\eta"'] \\
T & T & T
\end{tikzcd}
\tag{1}$$